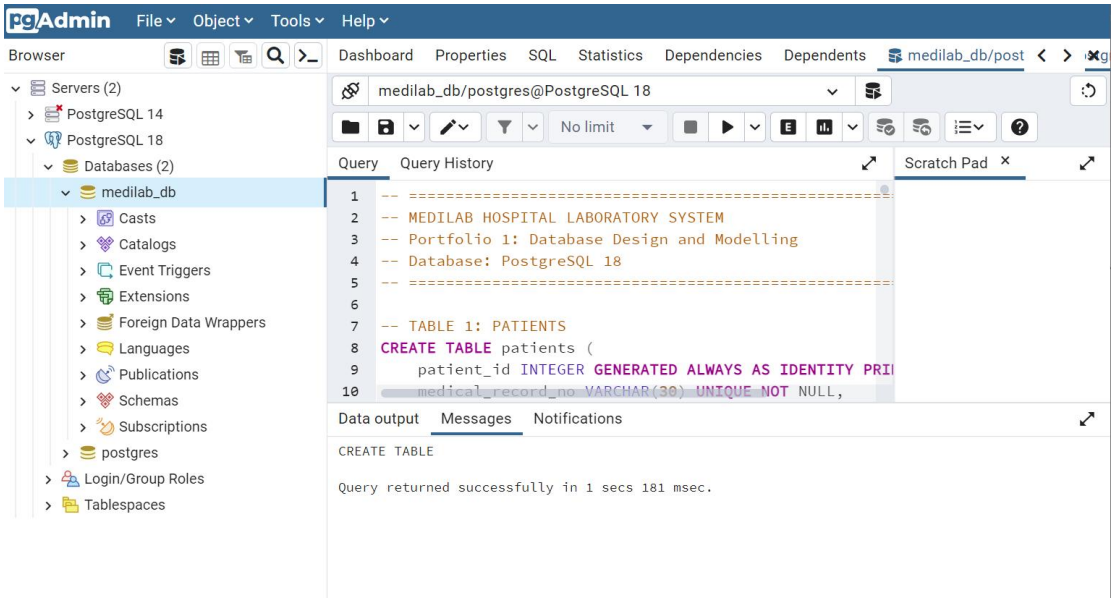
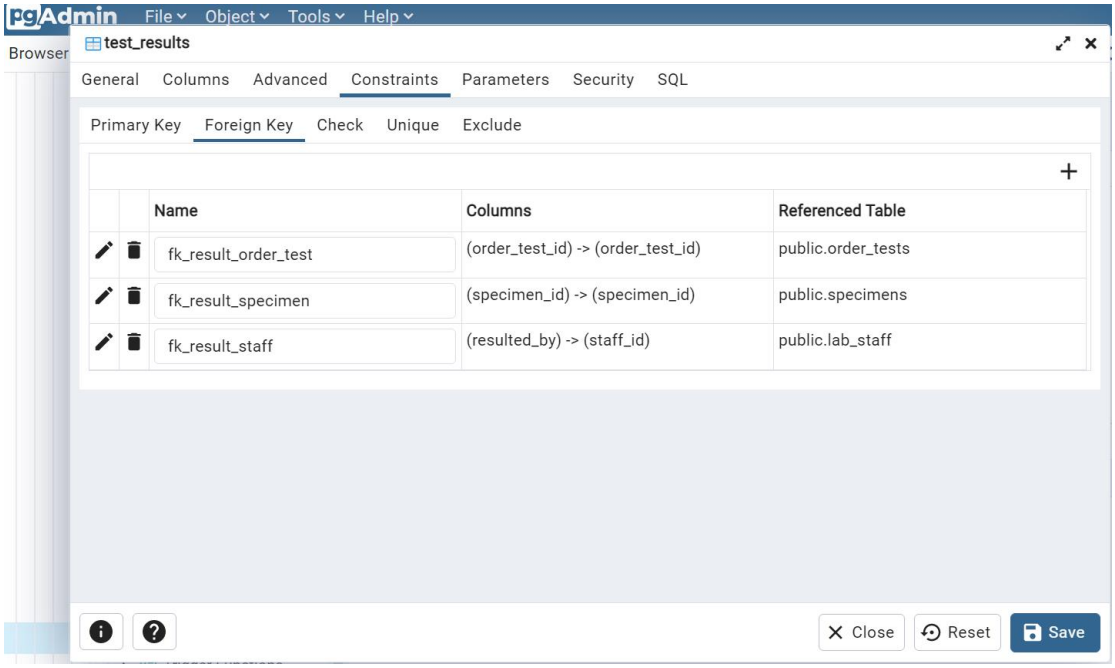
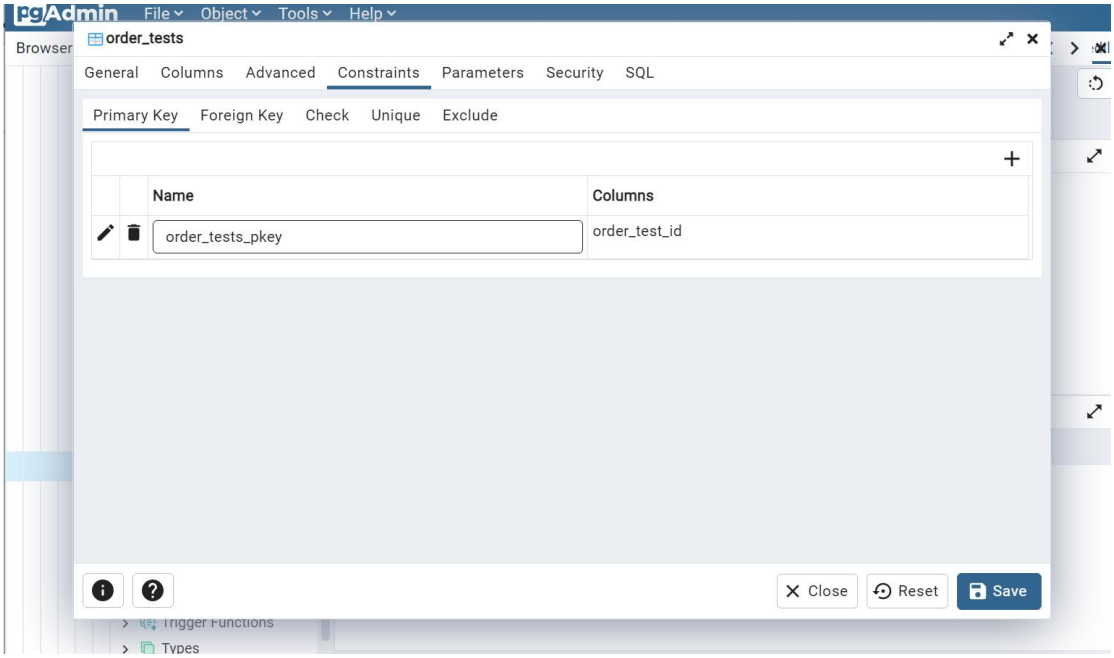
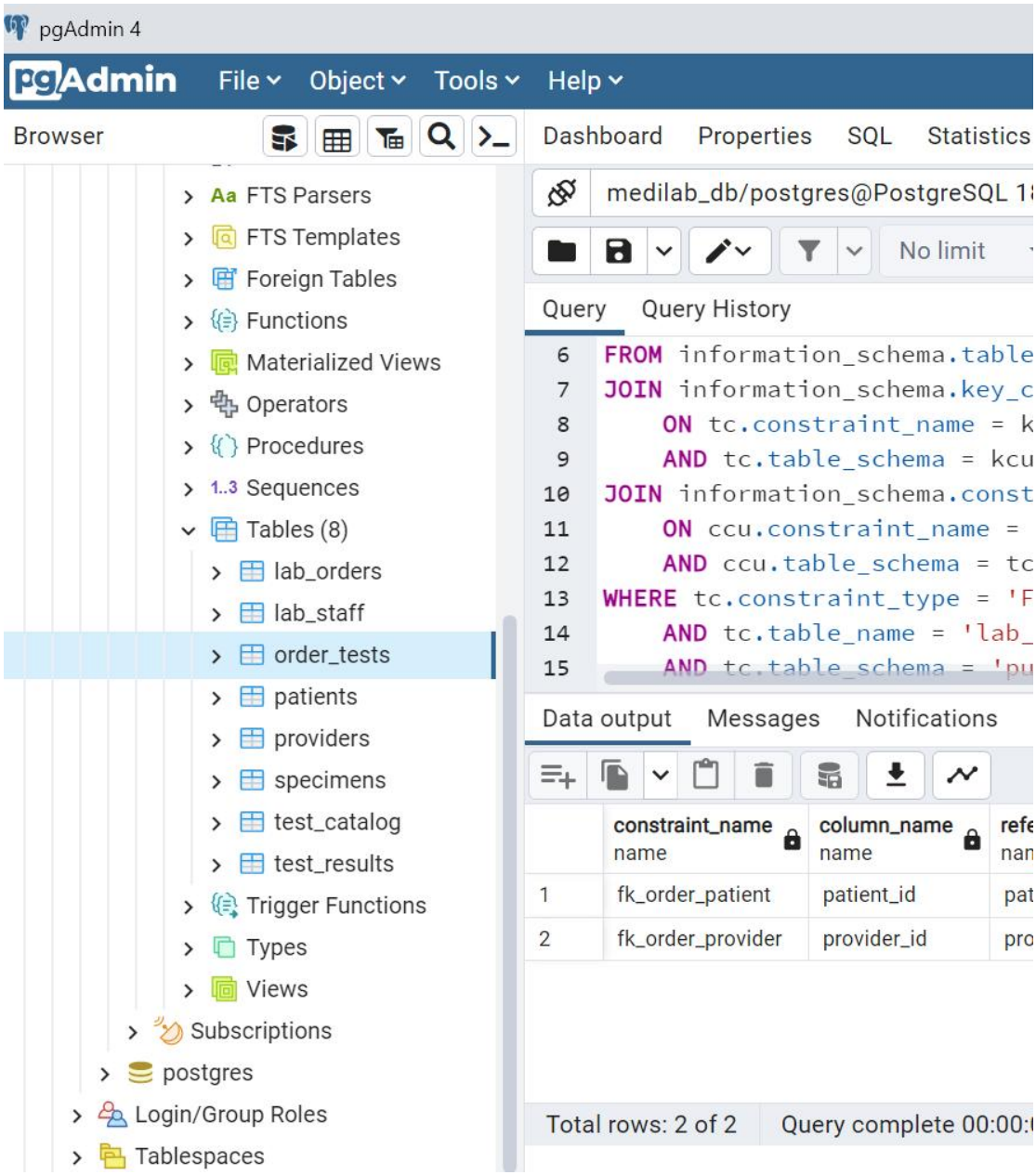
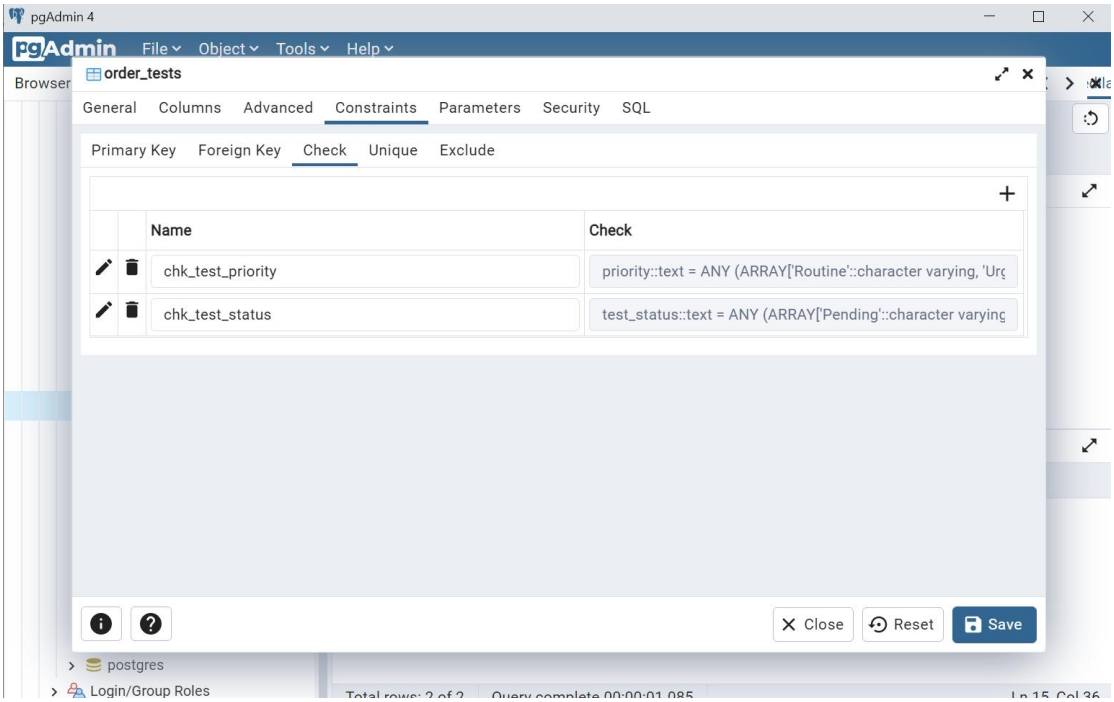
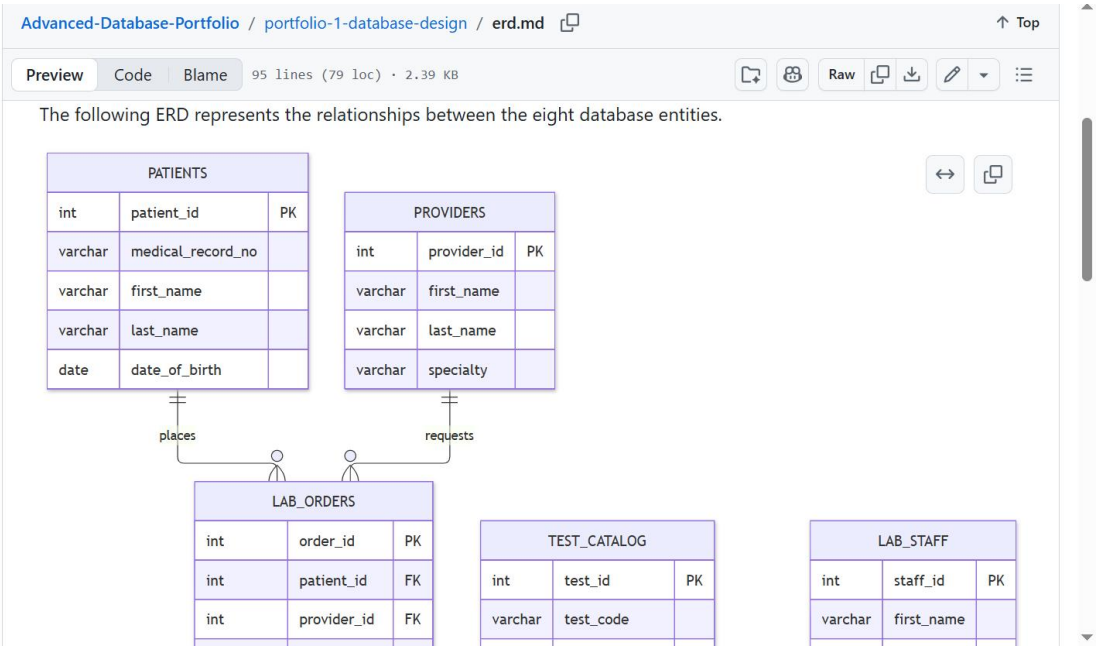

pgAdmin
File
Object
Tools
Help
Browser
Dashboard
Properties
SQL
Statistics
Dependencies
Dependents
medilab_db/post
Servers (2)
PostgreSQL 14
PostgreSQL 18
Databases (2)
medilab_db
Casts
Catalogs
Event Triggers
Extensions
Foreign Data Wrappers
Languages
Publications
Schemas
Subscriptions
postgres
Login/Group Roles
Tablespaces
medilab_db/postgres@PostgreSQL 18
No limit
Query
Query History
Scratch Pad
1 -- ============================================================
2 -- MEDILAB HOSPITAL LABORATORY SYSTEM
3 -- Portfolio 1: Database Design and Modelling
4 -- Database: PostgreSQL 18
5 -- ============================================================
6
7 -- TABLE 1: PATIENTS
8 CREATE TABLE patients (
9 patient_id INTEGER GENERATED ALWAYS AS IDENTITY PRI
10 medical_record_no VARCHAR(30) UNIQUE NOT NULL,
Data output
Messages
Notifications
CREATE TABLE
Query returned successfully in 1 secs 181 msec.

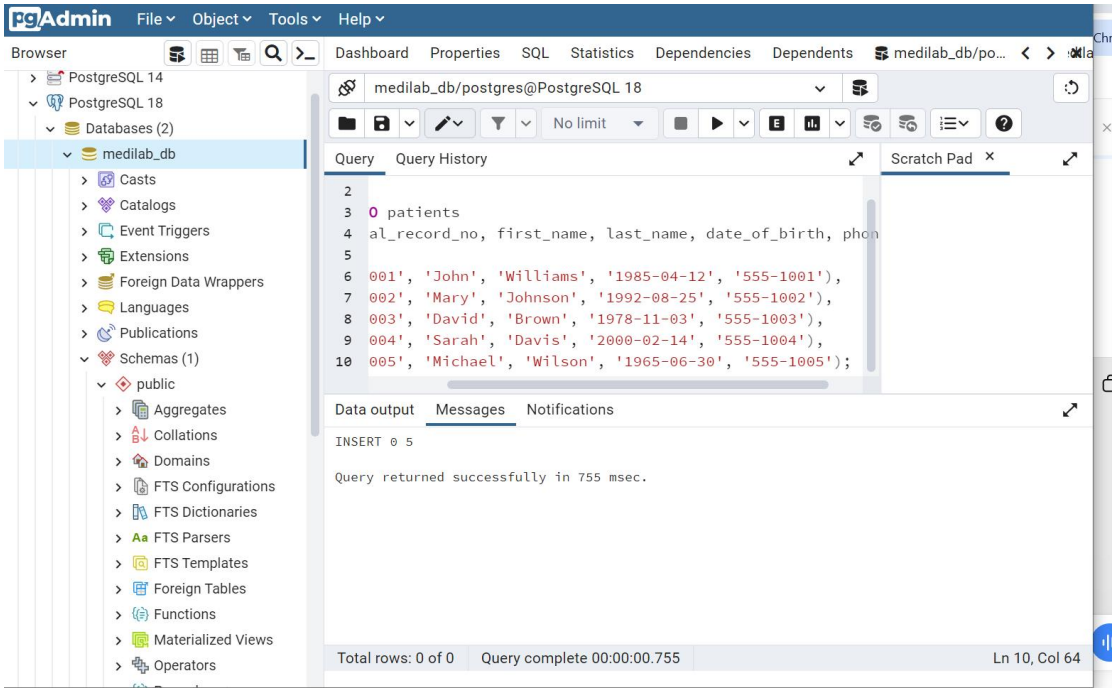

pgAdmin
File
Object
Tools
Help
Browser
test_results
General
Columns
Advanced
Constraints
Parameters
Security
SQL
Primary Key
Foreign Key
Check
Unique
Exclude
Name
Columns
Referenced Table
fk_result_order_test
(order_test_id) -> (order_test_id)
public.order_tests
fk_result_specimen
(specimen_id) -> (specimen_id)
public.specimens
fk_result_staff
(resulted_by) -> (staff_id)
public.lab_staff
Close
Reset
Save

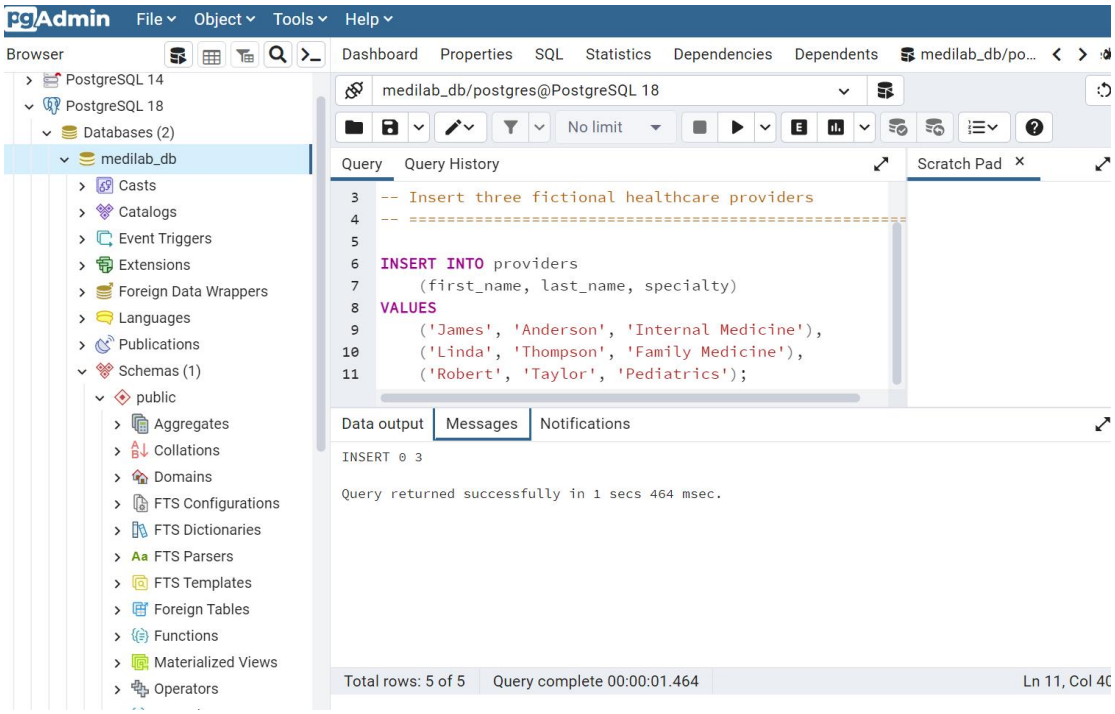

pgAdmin
File
Object
Tools
Help
Browser
order_tests
General
Columns
Advanced
Constraints
Parameters
Security
SQL
Primary Key
Foreign Key
Check
Unique
Exclude
Name
Columns
order_tests_pkey
order_test_id
Close
Reset
Save
Trigger Functions
Types

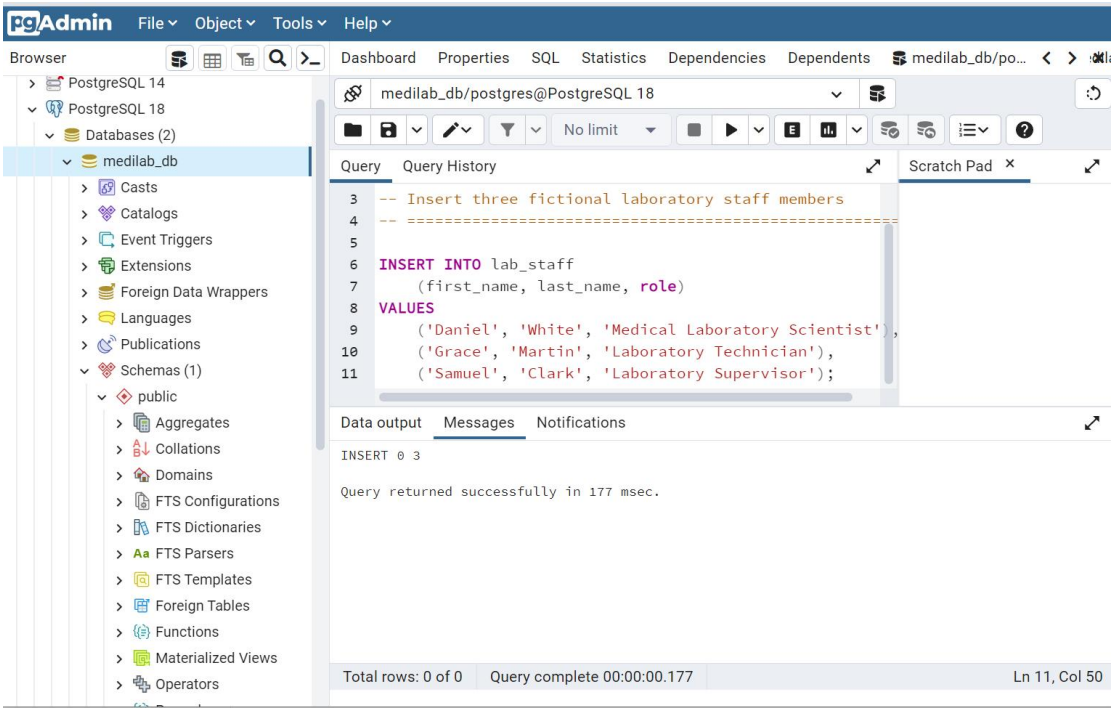

pgAdmin 4
pgAdmin
File
Object
Tools
Help
Browser
Dashboard
Properties
SQL
Statistics
FTS Parsers
FTS Templates
Foreign Tables
Functions
Materialized Views
Operators
Procedures
Sequences
Tables (8)
lab_orders
lab_staff
order_tests
patients
providers
specimens
test_catalog
test_results
Trigger Functions
Types
Views
Subscriptions
postgres
Login/Group Roles
Tablespaces
medilab_db/postgres@PostgreSQL 1
No limit
Query
Query History
FROM information_schema.table
JOIN information_schema.key_c
ON tc.constraint_name = k
AND tc.table_schema = kcu
JOIN information_schema.const
ON ccu.constraint_name =
AND ccu.table_schema = tc
WHERE tc.constraint_type = 'F
AND tc.table_name = 'lab_
AND tc.table_schema = 'pu
Data output
Messages
Notifications
constraint_name
name
column_name
name
1
fk_order_patient
patient_id
2
fk_order_provider
provider_id
Total rows: 2 of 2
Query complete 00:00:

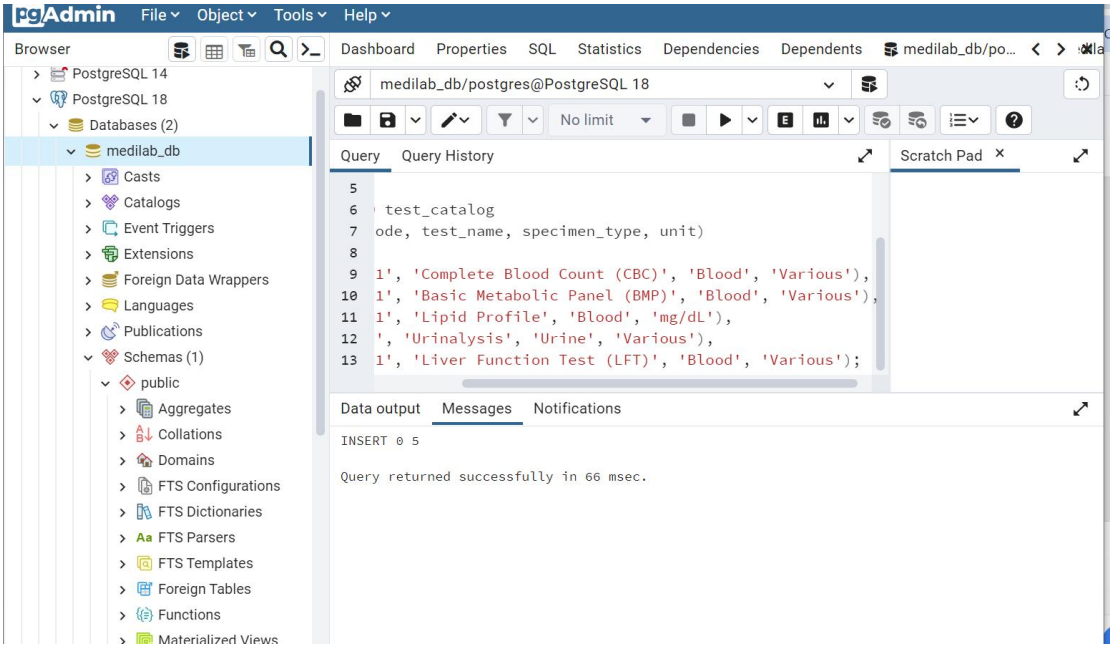

pgAdmin 4
File
Object
Tools
Help
Browser
order_tests
General
Columns
Advanced
Constraints
Parameters
Security
SQL
Primary Key
Foreign Key
Check
Unique
Exclude
Name
Check
chk_test_priority
priority::text = ANY (ARRAY['Routine'::character varying, 'Urg
chk_test_status
test_status::text = ANY (ARRAY['Pending'::character varying
Close
Reset
Save
postgres
Login/Group Roles

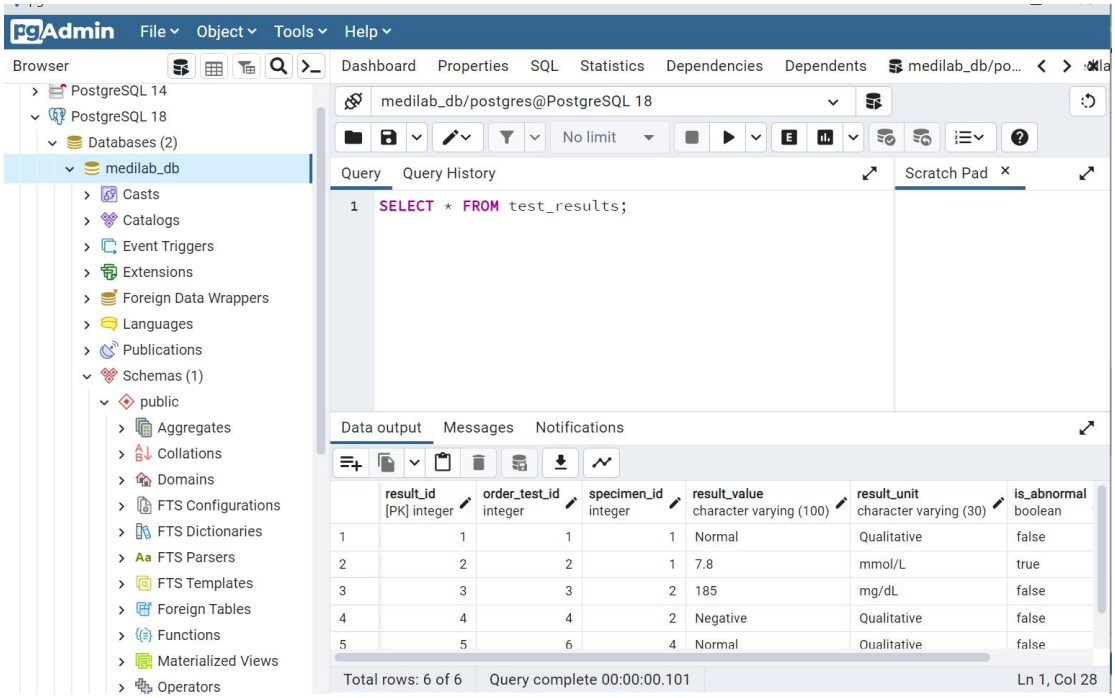

Advanced-Database-Portfolio / portfolio-1-database-design / erd.md
Top
Preview
Code
Blame
95 lines (79 loc) · 2.39 KB
Raw
The following ERD represents the relationships between the eight database entities.
PATIENTS
int patient_id PK
varchar medical_record_no
varchar first_name
varchar last_name
date date_of_birth
PROVIDERS
int provider_id PK
varchar first_name
varchar last_name
varchar specialty
places
requests
LAB_ORDERS
int order_id PK
int patient_id FK
int provider_id FK
TEST_CATALOG
int test_id PK
varchar test_code
LAB_STAFF
int staff_id PK
varchar first_name

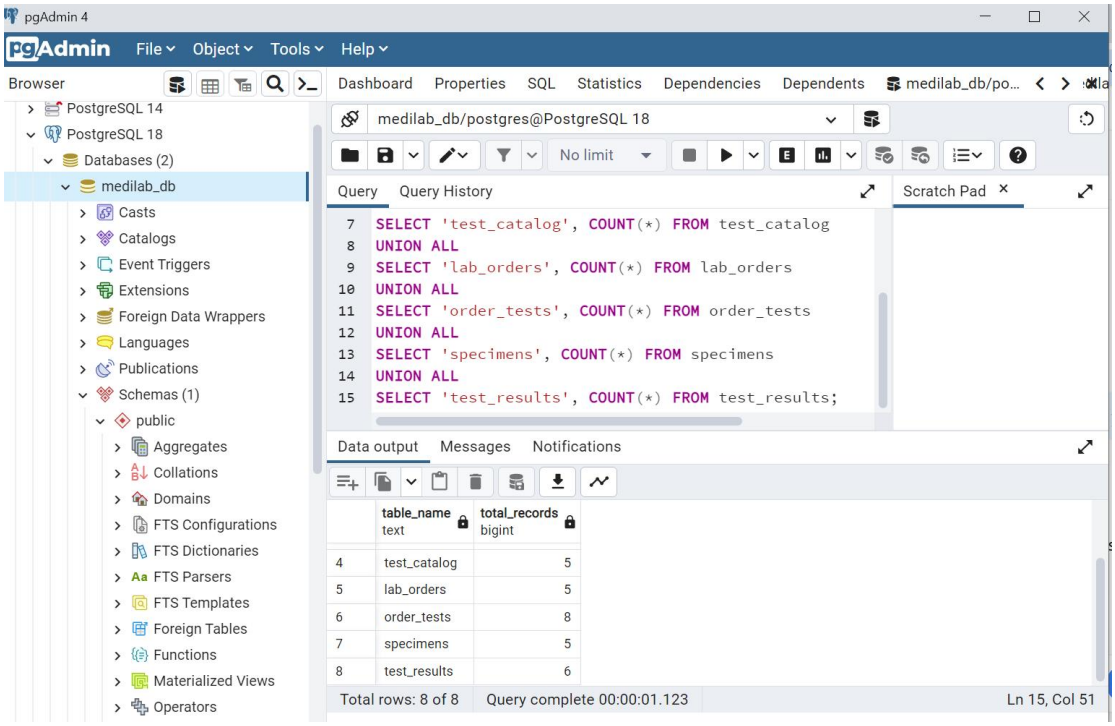

pgAdmin
File
Object
Tools
Help
Browser
Dashboard
Properties
SQL
Statistics
Dependencies
Dependents
medilab_db/po...
PostgreSQL 14
PostgreSQL 18
Databases (2)
medilab_db
Casts
Catalogs
Event Triggers
Extensions
Foreign Data Wrappers
Languages
Publications
Schemas (1)
public
Aggregates
Collations
Domains
FTS Configurations
FTS Dictionaries
FTS Parsers
FTS Templates
Foreign Tables
Functions
Materialized Views
Operators
medilab_db/postgres@PostgreSQL 18
No limit
Query
Query History
Scratch Pad
2
3 O patients
4 al_record_no, first_name, last_name, date_of_birth, phon
5
6 001', 'John', 'Williams', '1985-04-12', '555-1001'),
7 002', 'Mary', 'Johnson', '1992-08-25', '555-1002'),
8 003', 'David', 'Brown', '1978-11-03', '555-1003'),
9 004', 'Sarah', 'Davis', '2000-02-14', '555-1004'),
10 005', 'Michael', 'Wilson', '1965-06-30', '555-1005');
Data output
Messages
Notifications
INSERT 0 5
Query returned successfully in 755 msec.
Total rows: 0 of 0
Query complete 00:00:00.755
Ln 10, Col 64

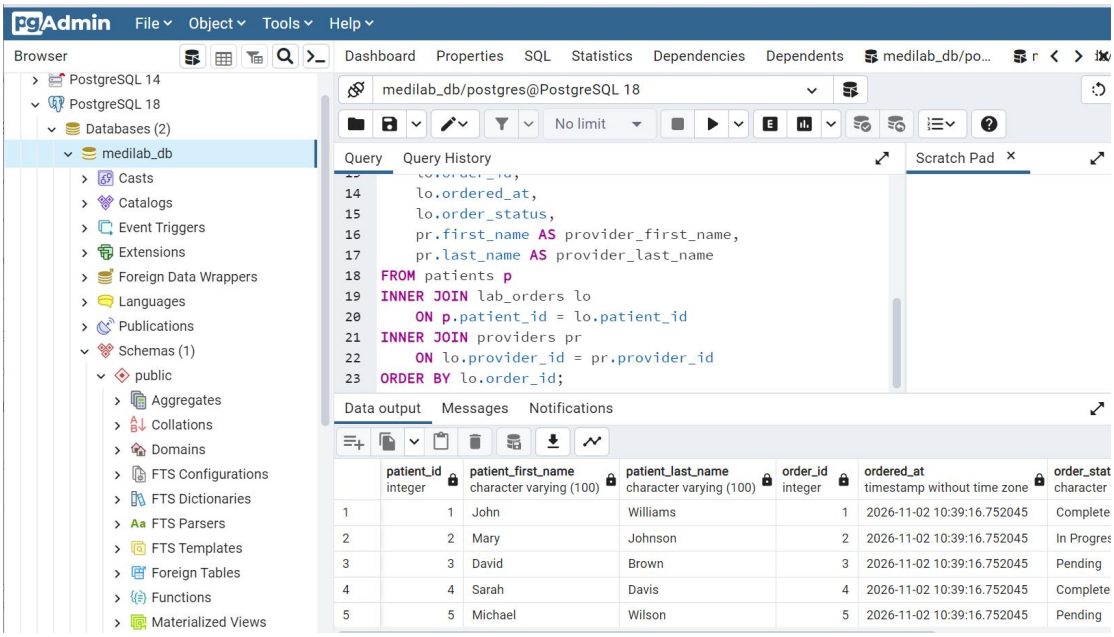

pgAdmin
File
Object
Tools
Help
Browser
Dashboard
Properties
SQL
Statistics
Dependencies
Dependents
medilab_db/po...
PostgreSQL 14
PostgreSQL 18
Databases (2)
medilab_db
Casts
Catalogs
Event Triggers
Extensions
Foreign Data Wrappers
Languages
Publications
Schemas (1)
public
Aggregates
Collations
Domains
FTS Configurations
FTS Dictionaries
FTS Parsers
FTS Templates
Foreign Tables
Functions
Materialized Views
Operators
medilab_db/postgres@PostgreSQL 18
No limit
Query
Query History
Scratch Pad
3 -- Insert three fictional healthcare providers
4 -- ==========
5
6 INSERT INTO providers
7 (first_name, last_name, specialty)
8 VALUES
9 ('James', 'Anderson', 'Internal Medicine'),
10 ('Linda', 'Thompson', 'Family Medicine'),
11 ('Robert', 'Taylor', 'Pediatrics');
Data output
Messages
Notifications
INSERT 0 3
Query returned successfully in 1 secs 464 msec.
Total rows: 5 of 5
Query complete 00:00:01.464
Ln 11, Col 40

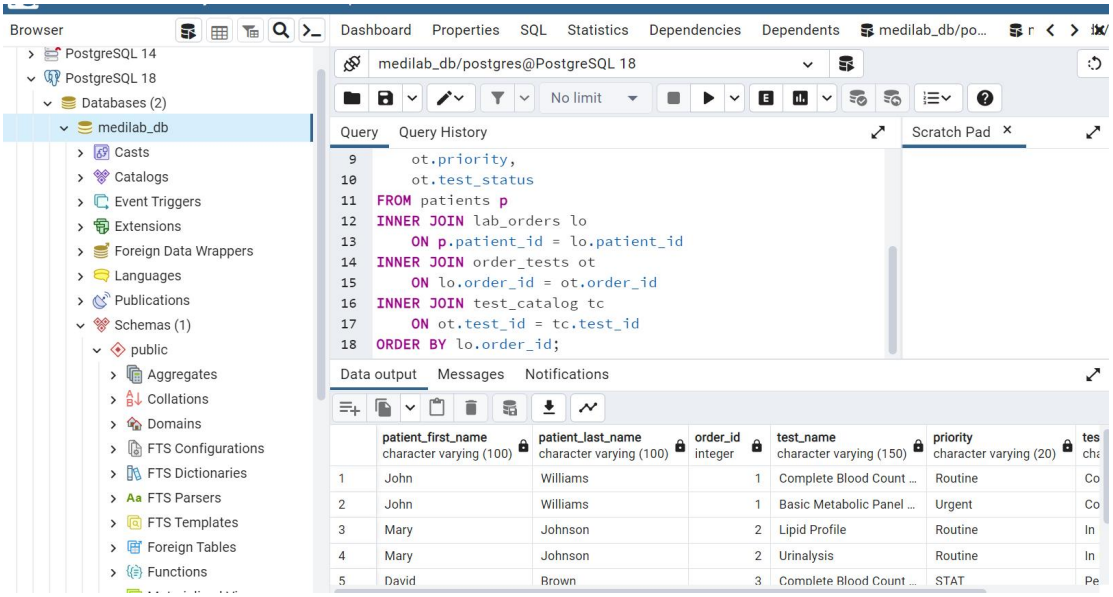

pgAdmin
File
Object
Tools
Help
Browser
Dashboard
Properties
SQL
Statistics
Dependencies
Dependents
medilab_db/po...
PostgreSQL 14
PostgreSQL 18
Databases (2)
medilab_db
Casts
Catalogs
Event Triggers
Extensions
Foreign Data Wrappers
Languages
Publications
Schemas (1)
public
Aggregates
Collations
Domains
FTS Configurations
FTS Dictionaries
FTS Parsers
FTS Templates
Foreign Tables
Functions
Materialized Views
Operators
medilab_db/postgres@PostgreSQL 18
No limit
Query
Query History
Scratch Pad
-- Insert three fictional laboratory staff members
INSERT INTO lab_staff
(first_name, last_name, role)
VALUES
('Daniel', 'White', 'Medical Laboratory Scientist'),
('Grace', 'Martin', 'Laboratory Technician'),
('Samuel', 'Clark', 'Laboratory Supervisor');
Data output
Messages
Notifications
INSERT 0 3
Query returned successfully in 177 msec.
Total rows: 0 of 0
Query complete 00:00:00.177
Ln 11, Col 50

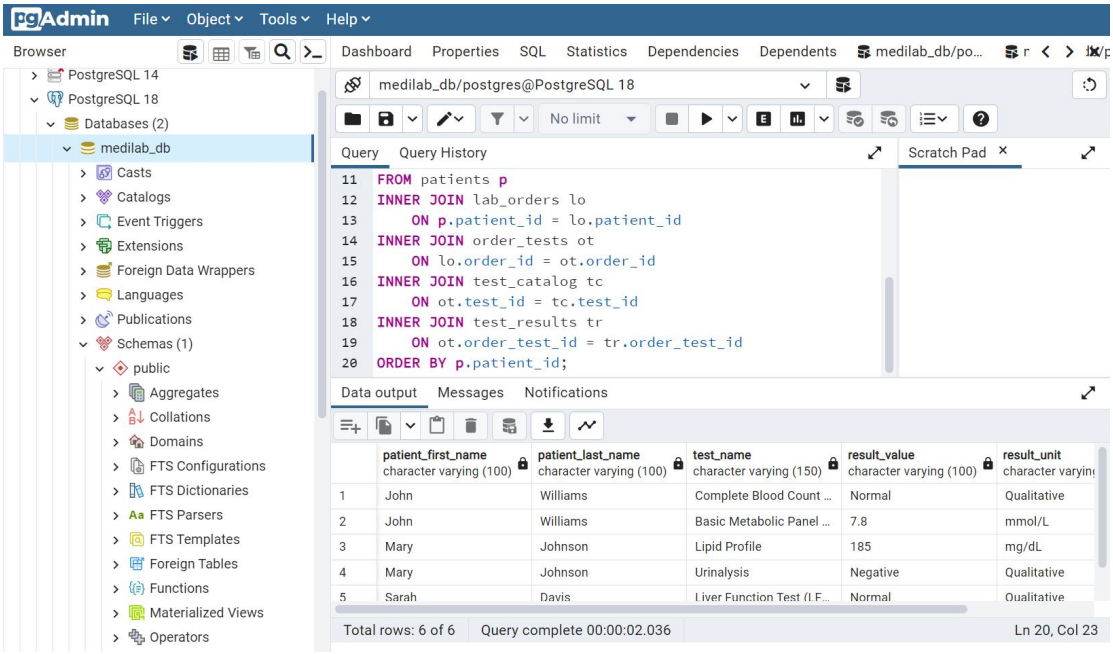

pgAdmin
File
Object
Tools
Help
Browser
Dashboard
Properties
SQL
Statistics
Dependencies
Dependents
medilab_db/po...
PostgreSQL 14
PostgreSQL 18
Databases (2)
medilab_db
Casts
Catalogs
Event Triggers
Extensions
Foreign Data Wrappers
Languages
Publications
Schemas (1)
public
Aggregates
Collations
Domains
FTS Configurations
FTS Dictionaries
FTS Parsers
FTS Templates
Foreign Tables
Functions
Materialized Views
medilab_db/postgres@PostgreSQL 18
No limit
Query
Query History
Scratch Pad
test_catalog
ode, test_name, specimen_type, unit)
1', 'Complete Blood Count (CBC)', 'Blood', 'Various'),
1', 'Basic Metabolic Panel (BMP)', 'Blood', 'Various'),
1', 'Lipid Profile', 'Blood', 'mg/dL'),
', 'Urinalysis', 'Urine', 'Various'),
1', 'Liver Function Test (LFT)', 'Blood', 'Various');
Data output
Messages
Notifications
INSERT 0 5
Query returned successfully in 66 msec.

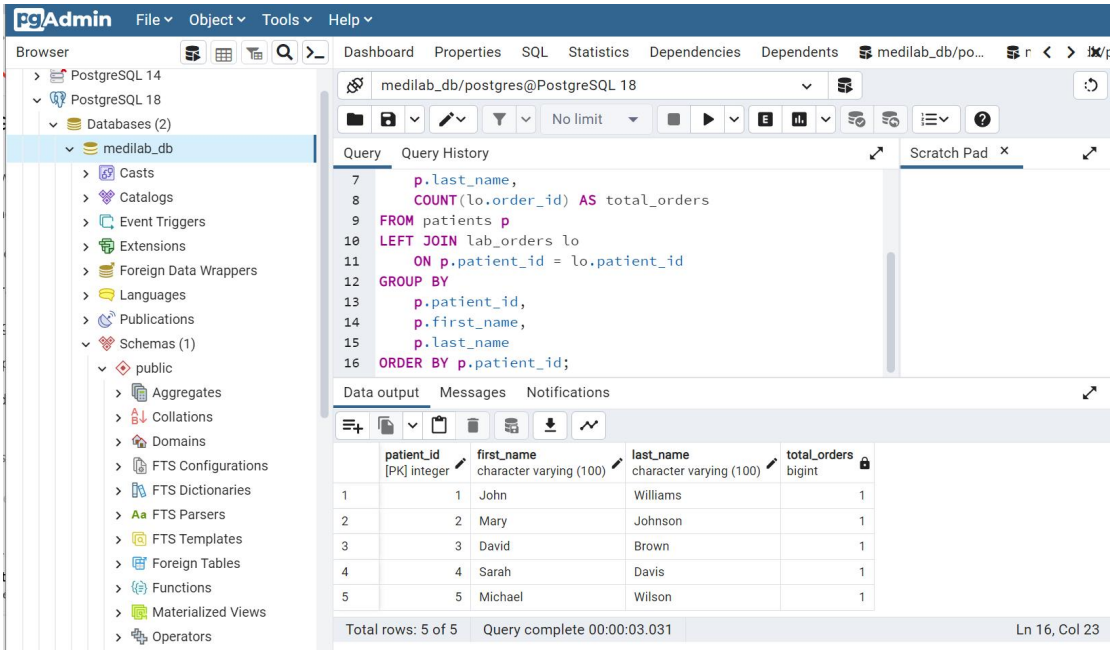

pgAdmin
File
Object
Tools
Help
Browser
Dashboard
Properties
SQL
Statistics
Dependencies
Dependents
medilab_db/po...
PostgreSQL 14
PostgreSQL 18
Databases (2)
medilab_db
Casts
Catalogs
Event Triggers
Extensions
Foreign Data Wrappers
Languages
Publications
Schemas (1)
public
Aggregates
Collations
Domains
FTS Configurations
FTS Dictionaries
FTS Parsers
FTS Templates
Foreign Tables
Functions
Materialized Views
Operators
medilab_db/postgres@PostgreSQL 18
No limit
Query
Query History
Scratch Pad
1 SELECT * FROM test_results;
Data output
Messages
Notifications
result_id [PK] integer
order_test_id integer
specimen_id integer
result_value character varying (100)
result_unit character varying (30)
is_abnormal boolean
1 1 1 1 Normal Qualitative false
2 2 2 1 7.8 mmol/L true
3 3 3 2 185 mg/dL false
4 4 4 2 Negative Qualitative false
5 5 6 4 Normal Qualitative false
Total rows: 6 of 6
Query complete 00:00:00.101
Ln 1, Col 28

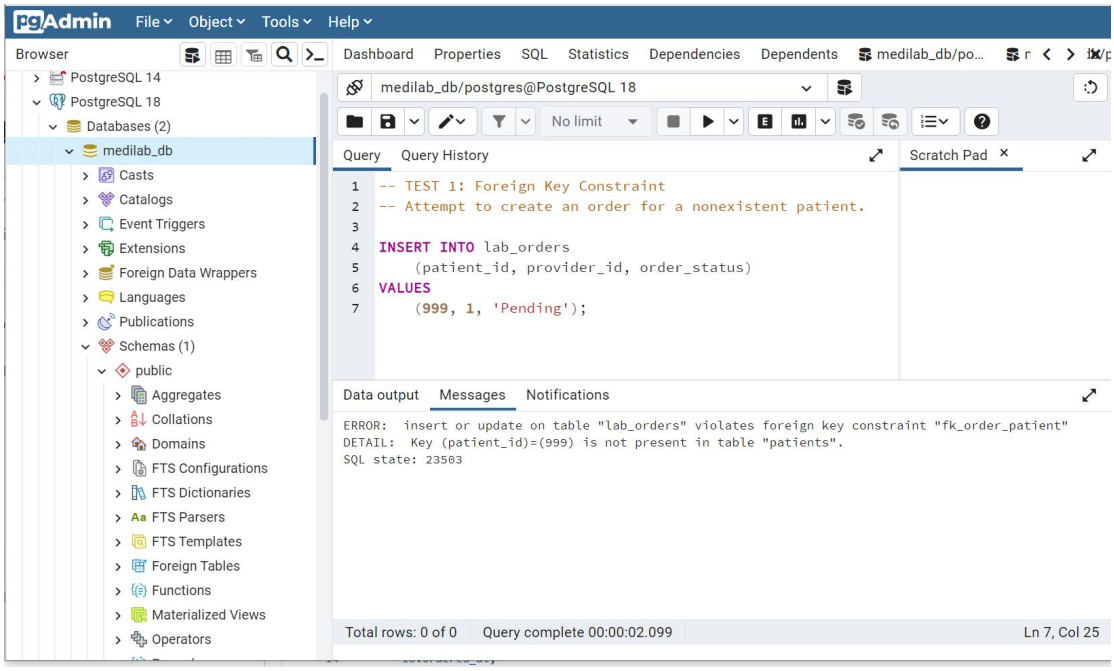

pgAdmin 4
pgAdmin
File
Object
Tools
Help
Browser
Dashboard
Properties
SQL
Statistics
Dependencies
Dependents
medilab_db/po...
PostgreSQL 14
PostgreSQL 18
Databases (2)
medilab_db
Casts
Catalogs
Event Triggers
Extensions
Foreign Data Wrappers
Languages
Publications
Schemas (1)
public
Aggregates
Collations
Domains
FTS Configurations
FTS Dictionaries
FTS Parsers
FTS Templates
Foreign Tables
Functions
Materialized Views
Operators
medilab_db/postgres@PostgreSQL 18
No limit
Query
Query History
Scratch Pad
7 SELECT 'test_catalog', COUNT(*) FROM test_catalog
8 UNION ALL
9 SELECT 'lab_orders', COUNT(*) FROM lab_orders
10 UNION ALL
11 SELECT 'order_tests', COUNT(*) FROM order_tests
12 UNION ALL
13 SELECT 'specimens', COUNT(*) FROM specimens
14 UNION ALL
15 SELECT 'test_results', COUNT(*) FROM test_results;
Data output
Messages
Notifications
table_name text
total_records bigint
4 test_catalog 5
5 lab_orders 5
6 order_tests 8
7 specimens 5
8 test_results 6
Total rows: 8 of 8
Query complete 00:00:01.123
Ln 15, Col 51

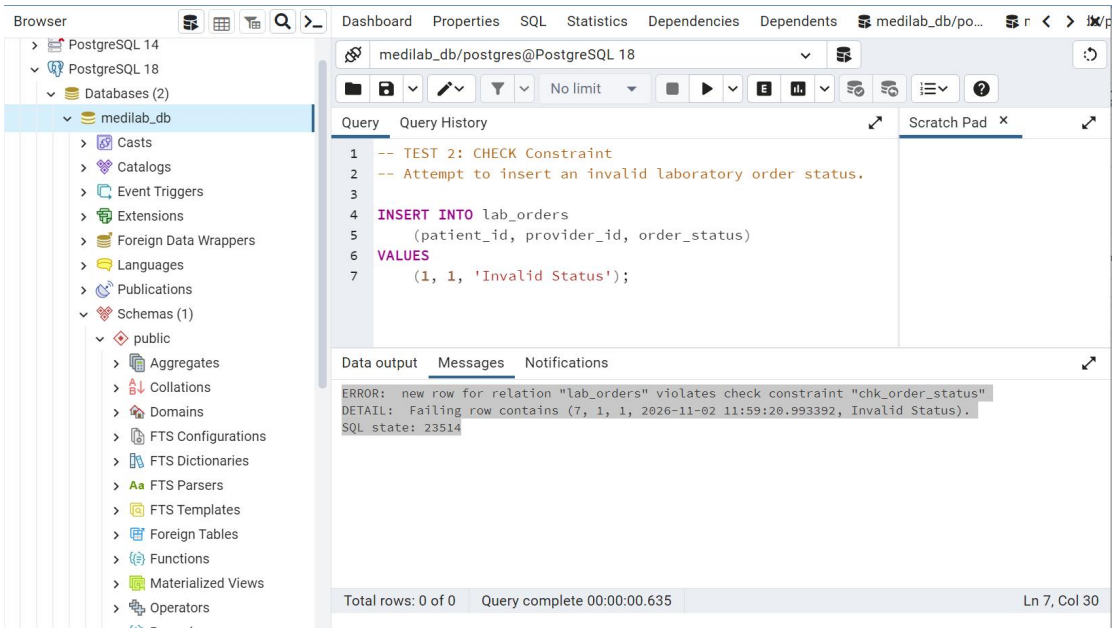

pgAdmin
File
Object
Tools
Help
Browser
Dashboard
Properties
SQL
Statistics
Dependencies
Dependents
medilab_db/postgres@PostgreSQL 18
No limit
Query
Query History
Scratch Pad
PostgreSQL 14
PostgreSQL 18
Databases (2)
medilab_db
Casts
Catalogs
Event Triggers
Extensions
Foreign Data Wrappers
Languages
Publications
Schemas (1)
public
Aggregates
Collations
Domains
FTS Configurations
FTS Dictionaries
FTS Parsers
FTS Templates
Foreign Tables
Functions
Materialized Views
14 lo.ordered_at,
15 lo.order_status,
16 pr.first_name AS provider_first_name,
17 pr.last_name AS provider_last_name
18 FROM patients p
19 INNER JOIN lab_orders lo
20 ON p.patient_id = lo.patient_id
21 INNER JOIN providers pr
22 ON lo.provider_id = pr.provider_id
23 ORDER BY lo.order_id;
Data output
Messages
Notifications
patient_id integer
patient_first_name character varying (100)
patient_last_name character varying (100)
order_id integer
ordered_at timestamp without time zone
order_stat character
1 1 John Williams 1 2026-11-02 10:39:16.752045 Complete
2 2 Mary Johnson 2 2026-11-02 10:39:16.752045 In Progres
3 3 David Brown 3 2026-11-02 10:39:16.752045 Pending
4 4 Sarah Davis 4 2026-11-02 10:39:16.752045 Complete
5 5 Michael Wilson 5 2026-11-02 10:39:16.752045 Pending


Browser
Dashboard
Properties
SQL
Statistics
Dependencies
Dependents
medilab_db/postgres@PostgreSQL 18
No limit
Query
Query History
Scratch Pad
PostgreSQL 14
PostgreSQL 18
Databases (2)
medilab_db
Casts
Catalogs
Event Triggers
Extensions
Foreign Data Wrappers
Languages
Publications
Schemas (1)
public
Aggregates
Collations
Domains
FTS Configurations
FTS Dictionaries
FTS Parsers
FTS Templates
Foreign Tables
Functions
9 ot.priority,
10 ot.test_status
11 FROM patients p
12 INNER JOIN lab_orders lo
13 ON p.patient_id = lo.patient_id
14 INNER JOIN order_tests ot
15 ON lo.order_id = ot.order_id
16 INNER JOIN test_catalog tc
17 ON ot.test_id = tc.test_id
18 ORDER BY lo.order_id;
Data output
Messages
Notifications
patient_first_name character varying (100)
patient_last_name character varying (100)
order_id integer
test_name character varying (150)
priority character varying (20)
1 John Williams 1 Complete Blood Count ... Routine
2 John Williams 1 Basic Metabolic Panel ... Urgent
3 Mary Johnson 2 Lipid Profile Routine
4 Mary Johnson 2 Urinalysis Routine
5 David Brown 3 Complete Blood Count ... STAT


pgAdmin
File
Object
Tools
Help
Browser
Dashboard
Properties
SQL
Statistics
Dependencies
Dependents
medilab_db/po...
PostgreSQL 14
PostgreSQL 18
Databases (2)
medilab_db
Casts
Catalogs
Event Triggers
Extensions
Foreign Data Wrappers
Languages
Publications
Schemas (1)
public
Aggregates
Collations
Domains
FTS Configurations
FTS Dictionaries
FTS Parsers
FTS Templates
Foreign Tables
Functions
Materialized Views
Operators
medilab_db/postgres@PostgreSQL 18
No limit
Query
Query History
Scratch Pad
11 FROM patients p
12 INNER JOIN lab_orders lo
13 ON p.patient_id = lo.patient_id
14 INNER JOIN order_tests ot
15 ON lo.order_id = ot.order_id
16 INNER JOIN test_catalog tc
17 ON ot.test_id = tc.test_id
18 INNER JOIN test_results tr
19 ON ot.order_test_id = tr.order_test_id
20 ORDER BY p.patient_id;
Data output
Messages
Notifications
patient_first_name character varying (100)
patient_last_name character varying (100)
test_name character varying (150)
result_value character varying (100)
result_unit character varyin
1 John Williams Complete Blood Count ... Normal Qualitative
2 John Williams Basic Metabolic Panel ... 7.8 mmol/L
3 Mary Johnson Lipid Profile 185 mg/dL
4 Mary Johnson Urinalysis Negative Qualitative
5 Sarah Davis Liver Function Test (LF... Normal Qualitative
Total rows: 6 of 6
Query complete 00:00:02.036
Ln 20, Col 23


pgAdmin
File
Object
Tools
Help
Browser
Dashboard
Properties
SQL
Statistics
Dependencies
Dependents
medilab_db/po...
PostgreSQL 14
PostgreSQL 18
Databases (2)
medilab_db
Casts
Catalogs
Event Triggers
Extensions
Foreign Data Wrappers
Languages
Publications
Schemas (1)
public
Aggregates
Collations
Domains
FTS Configurations
FTS Dictionaries
FTS Parsers
FTS Templates
Foreign Tables
Functions
Materialized Views
Operators
medilab_db/postgres@PostgreSQL 18
No limit
Query
Query History
Scratch Pad
7 p.last_name,
8 COUNT(lo.order_id) AS total_orders
9 FROM patients p
10 LEFT JOIN lab_orders lo
11 ON p.patient_id = lo.patient_id
12 GROUP BY
13 p.patient_id,
14 p.first_name,
15 p.last_name
16 ORDER BY p.patient_id;
Data output
Messages
Notifications
patient_id [PK] integer
first_name character varying (100)
last_name character varying (100)
total_orders bigint
1 1 John Williams 1
2 2 Mary Johnson 1
3 3 David Brown 1
4 4 Sarah Davis 1
5 5 Michael Wilson 1
Total rows: 5 of 5
Query complete 00:00:03.031
Ln 16, Col 23


pgAdmin
File
Object
Tools
Help
Browser
Dashboard
Properties
SQL
Statistics
Dependencies
Dependents
medilab_db/po...
PostgreSQL 14
PostgreSQL 18
Databases (2)
medilab_db
Casts
Catalogs
Event Triggers
Extensions
Foreign Data Wrappers
Languages
Publications
Schemas (1)
public
Aggregates
Collations
Domains
FTS Configurations
FTS Dictionaries
FTS Parsers
FTS Templates
Foreign Tables
Functions
Materialized Views
Operators
medilab_db/postgres@PostgreSQL 18
No limit
Query
Query History
Scratch Pad
1 -- TEST 1: Foreign Key Constraint
2 -- Attempt to create an order for a nonexistent patient.
3
4 INSERT INTO lab_orders
5 (patient_id, provider_id, order_status)
6 VALUES
7 (999, 1, 'Pending');
Data output
Messages
Notifications
ERROR: insert or update on table "lab_orders" violates foreign key constraint "fk_order_patient"
DETAIL: Key (patient_id)=(999) is not present in table "patients".
SQL state: 23503
Total rows: 0 of 0
Query complete 00:00:02.099
Ln 7, Col 25


Browser
Dashboard
Properties
SQL
Statistics
Dependencies
Dependents
medilab_db/po...
PostgreSQL 14
PostgreSQL 18
Databases (2)
medilab_db
Casts
Catalogs
Event Triggers
Extensions
Foreign Data Wrappers
Languages
Publications
Schemas (1)
public
Aggregates
Collations
Domains
FTS Configurations
FTS Dictionaries
FTS Parsers
FTS Templates
Foreign Tables
Functions
Materialized Views
Operators
medilab_db/postgres@PostgreSQL 18
No limit
Query
Query History
Scratch Pad
1 -- TEST 2: CHECK Constraint
2 -- Attempt to insert an invalid laboratory order status.
3
4 INSERT INTO lab_orders
5 (patient_id, provider_id, order_status)
6 VALUES
7 (1, 1, 'Invalid Status');
Data output
Messages
Notifications
ERROR: new row for relation "lab_orders" violates check constraint "chk_order_status"
DETAIL: Failing row contains (7, 1, 1, 2026-11-02 11:59:20.993392, Invalid Status).
SQL state: 23514
Total rows: 0 of 0
Query complete 00:00:00.635
Ln 7, Col 30